

22PH202 – Electromagnetism and Modern Physics\

Unit II – Magnetism

2.5 Properties of magnet

2.5.1 Magnetic Dipole and dipole moment

Magnetic dipoles are found to exist in magnetic materials (which are similar to electric dipoles in dielectric material). Magnetic dipoles may be thought of as small bar magnets composed of north and south poles instead of positive and negative electric charges.

Magnetic dipole, generally a tiny magnet of microscopic to subatomic dimensions, equivalent to a flow of electric charge around a loop. Electrons circulating around atomic nuclei, electrons spinning on their axes, and rotating positively charged atomic nuclei all are magnetic dipoles. The sum of these effects may cancel so that a given type of atom may not be a magnetic dipole. If they do not fully cancel, the atom is a permanent magnetic dipole, as are iron atoms.

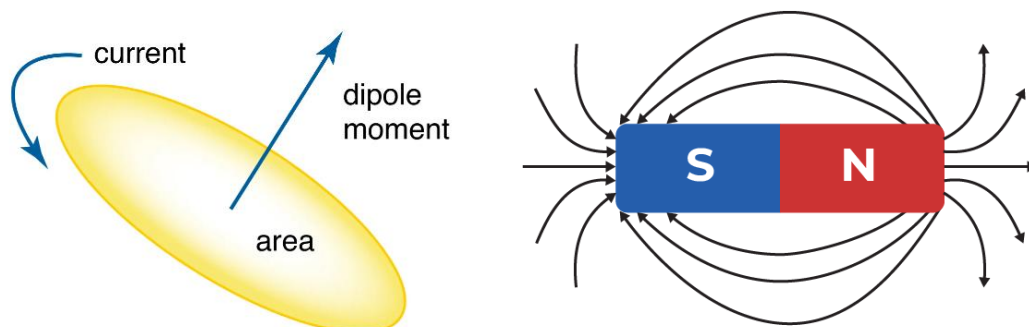


Figure 14. Magnetic dipole and poles of bar magnet

A magnetic moment is a quantity that represents the magnetic strength and orientation of a magnet or any other object that produces a magnetic field. The magnetic dipole moment is defined as the product of its pole strength and magnetic length. It is a vector quantity, denoted by P_m . The SI unit of magnetic moment is $A\ m^2$.

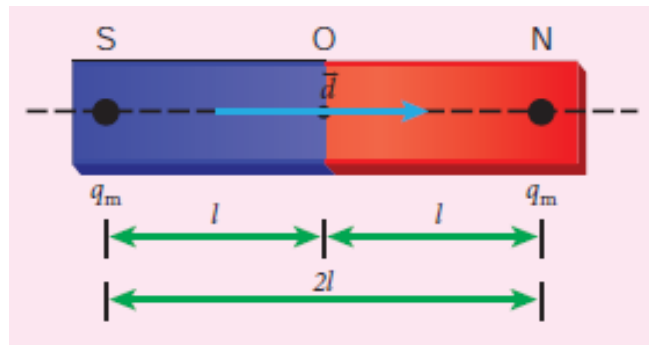


Figure 15. Magnetic dipole moment

$$P_m = 2lq_m \text{ (} q_m \text{: pole strength)}$$

2.6 Magnetic potential energy

Magnetic potential energy is energy stored in a system of magnets as a result of the exertion of a magnetic force on each other. When two magnets attract or repel each other, potential energy can be stored in their relative positions.

Magnetic potential energy is the energy stored because of attractive forces between two objects. For example, ball bearings, metal spoons, etc.,